

Vignette 4 | Microeconomics, Thursday, Sept. 30

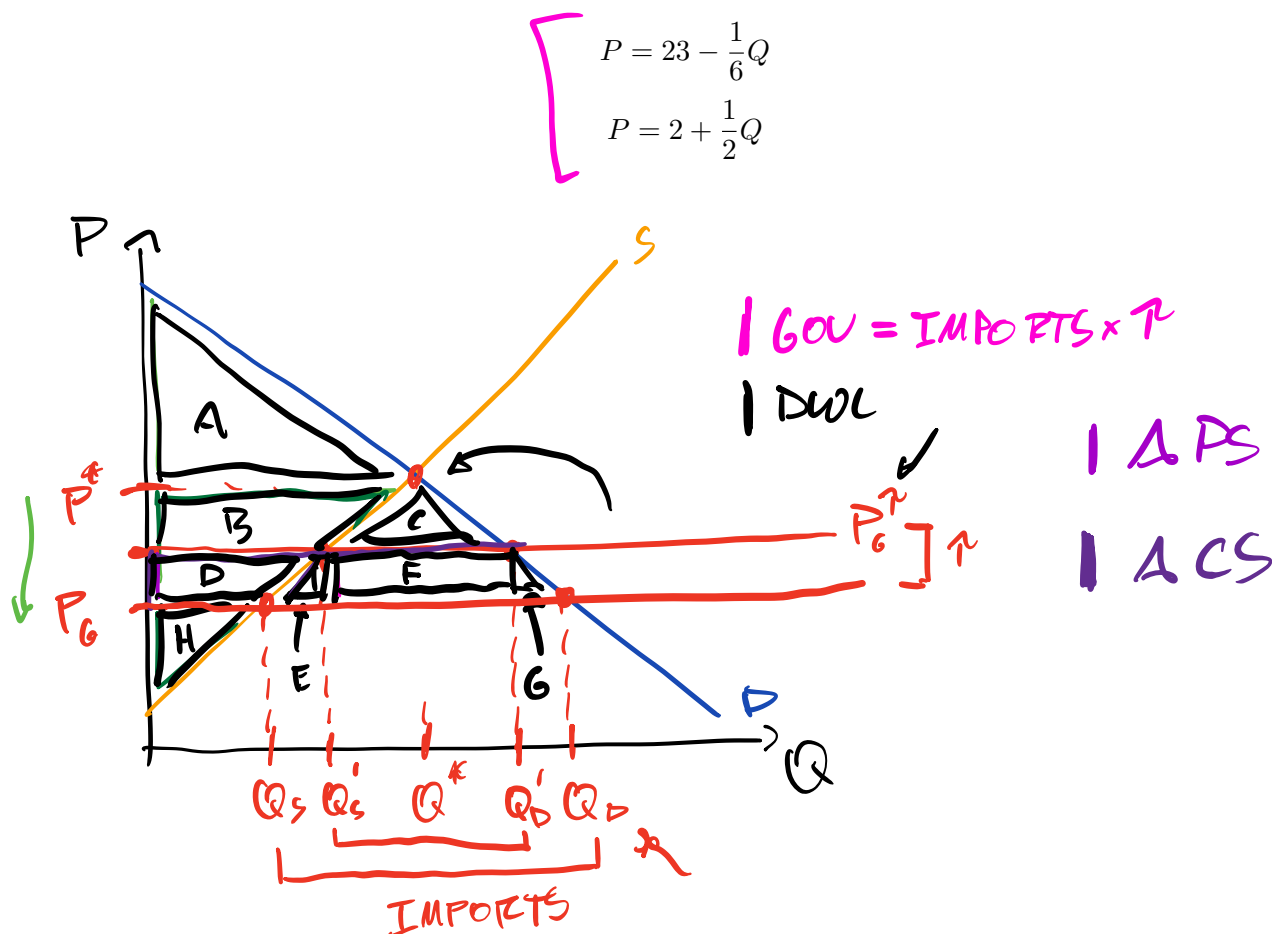
In response to anticipated political backlash associated with raising revenue via a tax on the much beloved butterbeer, the Ministry of Magic hired an external committee of advisors to find other means by which to raise government revenue. One classically trained magical economist proposed opening the closed domestic butterbeer market to the less expensive global market while imposing an import tariff. Ministry officials were pleased with the option but also raised concerns that domestic butterbeer sellers would be harmed by the lower price.

Use a graph to illustrate what would happen to the market after a tariff was imposed. Is there a tariff price that would alleviate the Ministry's concern about domestic sellers?

The supply and demand curves for butterbeer can be represented by the following equations.

$$P = 23 - \frac{1}{6}Q$$

$$P = 2 + \frac{1}{2}Q$$



PERIOD 1 | Domestic Market

$$CS = A$$

$$PS = B + D + H \quad \Delta PS_1$$

PERIOD 2 | Global Market

$$CS = A + B + C + D + E + F + G \quad \Delta CS_1$$

$$PS = H \quad \Delta CS_2$$

PERIOD 3 | Import Tariff

$$CS = A + B + C$$

$$PS = H + D \quad \Delta PS_2$$

$$GOV = F$$

$$DWL = E + G \quad \text{In Period 2 but not in Period 3}$$

PERIOD 1 $\rightarrow$ PERIOD 3 the change in PS is:

$$B + D + H \rightarrow H + D$$

After the opening and tariff, producers lost welfare equal to B .

The tariff mitigated the Ministry's concerns, but doesn't fully solve them.